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## Introduction

Dr. Lixiang Yan is an Assistant Professor at the Institute for Artificial Intelligence in Education, Tsinghua University. He specializes in the intersecting fields of Artificial Intelligence, Educational Technology, and Learning Analytics, with a research focus on the deep integration of cutting-edge technology and human learning processes. In recognition of his innovative work in multimodal learning analytics and the application of generative AI, he has been selected for the National High-Level Young Talent Program. Dr. Yan's research is centered on two core contribution areas: first, he pioneers the application of Multimodal Learning Analytics in complex collaborative learning environments to reveal deep-seated behavioral patterns and interaction mechanisms; second, he conducts systematic and forward-looking explorations into the potential and challenges of generative AI in education. His scholarly work has been published in top-tier journals such as *Nature Human Behaviour*, *Nature Reviews Psychology*, *Computers & Education*, and the *British Journal of Educational Technology*, exerting a broad and positive impact on the academic community.

## Education Background

- **Ph.D. in Information Technology**, 2020/07 - 2023/12
  - Monash University, Australia
- **Master of Applied Psychology**, 2018/07 - 2019/12
  - The University of Melbourne, Australia
- **Bachelor of Science in Psychology**, 2014/03 - 2016/12
  - The University of Melbourne, Australia

## Work Experience

- **Assistant Professor**, 2025/08 - Present
  - Tsinghua University
- **Postdoctoral Researcher**, 2023/12 - 2025/07
  - Monash University, Australia

## Professional Service

- **Guest Editor (Primary)**, *British Journal of Educational Technology*, 2026/03 - Now
- **Members At-Large**, *Society for Learning Analytics Research (SoLAR)*, 2026/02 - Now

- **Co-Chair (PIP Track)**, *The 27th International Conference on Artificial Intelligence in Education*, 2025/12 - 2026/07
- **Guest Editor (Primary)**, *British Journal of Educational Technology*, 2024/05 - 2025/09
- **Guest Editor**, *Journal of Learning Analytics*, 2024/05 - 2025/03
- **Early Career Researcher Editorial Board Member**, *British Journal of Educational Technology*, 2024/03 - Present
- **Workshop Organizer**, International Learning Analytics and Knowledge (LAK) Conference, 2024/03 - Present
- **Primary Summit Organizer**, Adaptive Skills and Knowledge for the Age of AI (ASK4AI) Melbourne Summit, 2024
- **Invited Guest Lecturer**, Faculty of Architecture, TU Wien in Vienna, Austria, 2024
- **Panel Member (Doctoral Level)**, Saleh Ramadhan Alghamdi, 2024
- **Panel Member (Doctoral Level)**, Sehrish Iqbal, 2024
- **Organizing Team (Student Representative)**, The Australian Learning Analytics Summer Institute (ALASI), 2022

## Teaching Program

- Educational Big Data and Learning Analytics (81030202-0)

## Research Field & Overview

- **Research Fields:** AI in Education, Generative AI, Educational Agents, Educational Technology, Multimodal Learning Analytics, Human-Computer Interaction, Collaborative Learning
- **Overview:** Dr. Yan's research is situated at the intersection of education and technology, aiming to provide insightful support for authentic learning and teaching scenarios using advanced theories and techniques from AI, data analytics, and HCI. His work is characterized by deep interdisciplinary integration and a strong connection between theory and practice.
  - **Contribution 1: Multimodal Learning Analytics in Collaborative Learning.** Dr. Yan has pioneered the use of Multimodal Learning Analytics (MMLA), integrating multi-source data (e.g., positioning, audio, physiological) to deeply investigate behavioral patterns, temporal dynamics, and influencing factors in collaborative learning, especially in complex contexts like healthcare simulation. This work provides a solid empirical basis for optimizing the design and evaluation of collaborative learning.
  - **Contribution 2: Generative AI in Education.** Dr. Yan has conducted systematic and forward-thinking research on the application of Generative AI in education. He has comprehensively examined its opportunities and challenges as a learning tool, exploring its potential to transform learning delivery and assessment while also analyzing associated model-related, ethical, and evaluation issues. He has led international academic discourse in this area by organizing special issues in top-tier journals.

## Research Projects

1. **Principal Investigator**, *Evaluating and Cultivating Generative AI Literacy in Computer Science Education*, FIT Early Career Researcher Seed Grant, Monash University (A\$23,000), 2024-2025.
2. **Principal Investigator**, *Augmenting Learning Analytics Dashboard with Interactive AI Agents*, OpenAI Research Grant, OpenAI (US\$18,000), 2024-2025.
3. **Core Project Member**, *Assessments for writing with generative artificial intelligence*, Australian Research Council (A\$700,000), 2024-2027.
4. **Chief Investigator**, *Empowering Indigenous Language Learning through Co-created Knowledge Graph*, Monash Data Futures Institute (A\$32,000), 2023-2024.
5. **Principal Investigator**, *Clatics - Real-time Classroom Analytics*, Monash Generator Accelerator Program, Monash University (A\$10,000), 2022-2023.
6. **Core Project Member**, *Human-centred Teamwork Analytics*, Australian Research Council (A\$240,000), 2021-2024.
7. **Core Project Member**, *Researching Digital Citizenship in Asia-Pacific*, UNESCO Grant (A\$39,000), 2021-2022.

### Fellowship, Honors and Awards

- **National High-Level Young Talent Program**, 2025
- **Best Paper Award Nomination**, The 15th International Conference on Learning Analytics & Knowledge (LAK'25), 2025.
- **Wiley Top Cited Article**, *British Journal of Educational Technology*, 2023-2024.
- **Best Paper Award Nomination**, The 14th International Conference on Learning Analytics & Knowledge (LAK'24), 2024 (2 papers).
- **Best Paper Award Nomination**, The 24th International Conference on Artificial Intelligence in Education (AIED'23), 2023.
- **Best Paper Award**, The 12th International Conference on Learning Analytics & Knowledge (LAK'22), 2022.
- **Wiley Top Cited Article**, *British Journal of Educational Technology*, 2021-2022.

### Invited Talks

1. **Invited Keynote Speaker**, *Generative AI and Learning Analytics*, The 14th International Learning Analytics and Knowledge Conference (LAK'24), Kyoto, Japan, 2024.
2. **Invited Webinar Speaker**, *Socio-spatial Learning Analytics for Embodied Collaborative Learning*, Society for Learning Analytics Research (SoLAR), Online, 2023.
3. **Invited Keynote Speaker**, *Practical and ethical challenges of large language models in education*, AIED for the Future International Forum & the BNU Digital Learning Festival, The College of Education for the Future at BNUZ, Online, 2023.
4. **Invited Symposium Speaker**, *How do teachers use open learning spaces? Mapping from teachers' socio-spatial data to spatial pedagogy*, The 20th Biennial EARLI Conference, Thessaloniki, Greece, 2023.

### Publications

Yan, L., Greiff, S., Lodge, J. M., & Gašević, D. (2025). Distinguishing performance gains from learning when using generative AI. *Nature Reviews Psychology*, 1-2.

Yan, L., Greiff, S., Teuber, Z., & Gašević, D. (2024). Promises and challenges of generative artificial intelligence for human learning. *Nature Human Behaviour*, 8(10), 1839-1850.

Yan, L., Martinez-Maldonado, R., Jin, Y., Echeverria, V., Milesi, M., Fan, J., . . . Gašević, D. (2025). The effects of generative AI agents and scaffolding on enhancing students' comprehension of visual learning analytics. *Computers & Education*.

Yan, L., Echeverria, V., Jin, Y., Fernandez-Nieto, G., Zhao, L., Li, X., . . . Martinez-Maldonado, R. (2024). Evidence-based multimodal learning analytics for feedback and reflection in collaborative learning. *British Journal of Educational Technology*, 55(5), 1900-1925.

Yan, L., Sha, L., Zhao, L., Li, Y., Martinez-Maldonado, R., Chen, G., . . . Gašević, D. (2023). Practical and ethical challenges of large language models in education: A systematic scoping review. *British Journal of Educational Technology*, 1-23.

Yan, L., Martinez-Maldonado, R., Zhao, L., Dix, S., Jaggard, H., Wotherspoon, R., . . . Gašević, D. (2023). The role of indoor positioning analytics in assessment of simulation-based learning. *British Journal of Educational Technology*, 54(1), 267-292.

Yan, L., Martinez-Maldonado, R., Gallo Cordoba, B., Deppeler, J., Corrigan, D., & Gašević, D. (2022). Mapping from proximity traces to socio-spatial behaviours and student progression at the school. *British Journal of Educational Technology*, 53(6), 1645-1664.

Yan, L., Whitelock-Wainwright, A., Guan, Q., Wen, G., Gašević, D., & Chen, G. (2021). Students' experience of online learning during the COVID-19 pandemic: A province-wide survey study. *British Journal of Educational Technology*, 52(5), 2038-2057.

Yan, L., Martinez-Maldonado, R., Swiecki, Z., Zhao, L., Li, X., & Gašević, D. (2024). Dissecting the temporal dynamics of embodied collaborative learning using multimodal learning analytics. *Journal of Educational Psychology*.

Yan, L., Gašević, D., Echeverria, V., Zhao, L., Jin, Y., Li, X., & Martinez-Maldonado, R. (2025). In Sync or Out of Sync? Understanding Stress and Learning Performance in Collaborative Healthcare Simulations through Physiological Synchrony and Arousal. *International Journal of Artificial Intelligence in Education*, 1-32.

Yan, L., Martinez-Maldonado, R., Cordoba, B. G., Deppeler, J., Corrigan, D., Nieto, G. F., & Gasevic, D. (2021). Footprints at school: Modelling in-class social dynamics from students' physical positioning traces. Paper presented at the LAK21: 11th international learning analytics and knowledge conference.

Yan, L., Gasevic, D., Echeverria, V., Jin, Y., Zhao, L., & Martinez-Maldonado, R. (2025). From Complexity to Parsimony: Integrating Latent Class Analysis to Uncover Multimodal Learning Patterns in

Collaborative Learning. Paper presented at the Proceedings of the 15th International Learning Analytics and Knowledge Conference.

Yan, L., Martinez-Maldonado, R., & Gasevic, D. (2024). Generative artificial intelligence in learning analytics: Contextualising opportunities and challenges through the learning analytics cycle. Paper presented at the Proceedings of the 14th Learning Analytics and Knowledge Conference.

Yan, L., Martinez-Maldonado, R., Zhao, L., Deppeler, J., Corrigan, D., & Gasevic, D. (2022). How do Teachers Use Open Learning Spaces? Mapping from Teachers' Socio-spatial Data to Spatial Pedagogy. Paper presented at the LAK22: 12th international learning analytics and knowledge conference.

Yan, L., Martinez-Maldonado, R., Zhao, L., Li, X., & Gasevic, D. (2023). SeNA: Modelling socio-spatial analytics on homophily by integrating social and epistemic network analysis. Paper presented at the LAK23: 13th International Learning Analytics and Knowledge Conference.

Yan, L., Martinez-Maldonado, R., Zhao, L., Li, X., & Gašević, D. (2023). Physiological synchrony and arousal as indicators of stress and learning performance in embodied collaborative learning. Paper presented at the International Conference on Artificial Intelligence in Education.

Yan, L., Talic, S., Wild, H., Gasevic, D., Gašević, D., Ilic, D., . . . Trauer, J. (2022). Transmission of SARS-CoV-2 in a primary school setting with and without public health measures using real-world contact data: A modelling study. *Journal of Global Health*, 12, 05034.

Yan, L., Tan, Y., Swiecki, Z., Gašević, D., Williamson Shaffer, D., Zhao, L., . . . Martinez-Maldonado, R. (2023). Characterising individual-level collaborative learning behaviours using ordered network analysis and wearable sensors. Paper presented at the International Conference on Quantitative Ethnography.

Yan, L., Zhao, L., Echeverria, V., Jin, Y., Alfredo, R., Li, X., . . . Martinez-Maldonado, R. (2024). VizChat: enhancing learning analytics dashboards with contextualised explanations using multimodal generative AI chatbots. Paper presented at the International Conference on Artificial Intelligence in Education.

Yan, L., Zhao, L., Gašević, D., Li, X., & Martinez-Maldonado, R. (2023). Socio-Spatial Learning Analytics in Co-located Collaborative Learning Spaces: A Systematic Literature Review. *Journal of Learning Analytics*, 10(3), 45-63.

Yan, L., Zhao, L., Gasevic, D., & Martinez-Maldonado, R. (2022). Scalability, Sustainability, and Ethicality of Multimodal Learning Analytics. Paper presented at the LAK22: 12th international learning analytics and knowledge conference.

Yan, L., Echeverria, V., Fernandez-Nieto, G. M., Jin, Y., Swiecki, Z., Zhao, L., . . . Martinez-Maldonado, R. (2024). Human-AI collaboration in thematic analysis using ChatGPT: A user study and design recommendations. Paper presented at the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems.

Yan, L. (2023). Socio-spatial Learning Analytics. Monash University,

Alfredo, R., Echeverria, V., Jin, Y., Yan, L., Swiecki, Z., Gašević, D., & Martinez-Maldonado, R. (2024). Human-centred learning analytics and AI in education: A systematic literature review. *Computers and Education: Artificial Intelligence*, 100215.

Alfredo, R., Echeverria, V., Zhao, L., Lawrence, L., Fan, J. X., Yan, L., . . . Martinez-Maldonado, R. (2024). Designing a human-centred learning analytics dashboard in-use. *Journal of Learning Analytics*, 11(3), 62-81.

Damşa, C., Echeverria, V., Popov, V., Chernikova, O., Karlgren, K., Milesi, M., . . . Yan, L. (2025). Enhancing Team-Based Medical Simulations: Learning Through Reflection with Analytics and AI Tools. Paper presented at the Proceedings of the 18th International Conference on Computer-Supported Collaborative Learning-CSCL 2025, pp. 516-524.

Echeverria, V., Martinez-Maldonado, R., Yan, L., Zhao, L., Fernandez-Nieto, G., Gašević, D., & Shum, S. B. (2022). HuCETA: A framework for human-centered embodied teamwork analytics. *IEEE Pervasive Computing*, 22(1), 39-49.

Echeverria, V., Yan, L., Zhao, L., Abel, S., Alfredo, R., Dix, S., . . . Buckingham Shum, S. (2024). TeamSlides: A multimodal teamwork analytics dashboard for teacher-guided reflection in a physical learning space. Paper presented at the Proceedings of the 14th learning analytics and knowledge conference.

Echeverria, V., Zhao, L., Alfredo, R., Milesi, M. E., Jin, Y., Abel, S., . . . Wotherspoon, R. (2025). TeamVision: An AI-powered Learning Analytics System for Supporting Reflection in Team-based Healthcare Simulation. Paper presented at the Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems.

Feng, S., Yan, L., Zhao, L., Maldonado, R. M., & Gašević, D. (2024). Heterogenous network analytics of small group teamwork: Using multimodal data to uncover individual behavioral engagement strategies. Paper presented at the Proceedings of the 14th learning analytics and knowledge conference.

Jin, Y., Echeverria, V., Yan, L., Zhao, L., Alfredo, R., Tsai, Y.-S., . . . Martinez-Maldonado, R. (2024). FATE in MMLA: A Student-Centred Exploration of Fairness, Accountability, Transparency, and Ethics in Multimodal Learning Analytics. *Journal of Learning Analytics*, 11(3), 6-23.

Jin, Y., Martinez-Maldonado, R., Gašević, D., & Yan, L. (2025). GLAT: The generative AI literacy assessment test. *Computers and Education: Artificial Intelligence*, 100436.

Jin, Y., Yan, L., Echeverria, V., Gašević, D., & Martinez-Maldonado, R. (2025). Generative AI in higher education: A global perspective of institutional adoption policies and guidelines. *Computers and Education: Artificial Intelligence*, 8, 100348.

Jin, Y., Yang, K., Yan, L., Echeverria, V., Zhao, L., Alfredo, R., . . . Gasevic, D. (2025). Chatting with a learning analytics dashboard: The role of generative AI literacy on learner interaction with conventional and scaffolding chatbots. Paper presented at the Proceedings of the 15th International Learning Analytics and Knowledge Conference.

Jovanovic, J., Gašević, D., Yan, L., Baker, G., Murray, A., & Gasevic, D. (2024). Explaining trace-based learner profiles with self-reports: The added value of psychological networks. *Journal of Computer Assisted Learning*, 40(4), 1481-1499.

Khosravi, H., Shibani, A., Jovanovic, J., Pardos, Z. A., & Yan, L. (2025). Generative AI and Learning Analytics: Pushing Boundaries, Preserving Principles. *Journal of Learning Analytics*, 12(1), 1-11.

Li, T., Yan, L., Iqbal, S., Srivastava, N., Singh, S., Raković, M., . . . Fan, Y. (2025). Analytics of self-regulated learning strategies and scaffolding: Associations with learning performance. *Computers and Education: Artificial Intelligence*, 100410.

Li, X., Yan, L., Zhao, L., Martinez-Maldonado, R., & Gasevic, D. (2023). CVPE: A computer vision approach for scalable and privacy-preserving socio-spatial, multimodal learning analytics. Paper presented at the LAK23: 13th International Learning Analytics and Knowledge Conference.

Li, Y., Sha, L., Yan, L., Lin, J., Raković, M., Galbraith, K., . . . Chen, G. (2023). Can large language models write reflectively. *Computers and Education: Artificial Intelligence*, 4, 100140.

Martinez-Maldonado, R., Echeverria, V., Fernandez-Nieto, G., Yan, L., Zhao, L., Alfredo, R., . . . Wotherspoon, R. (2023). Lessons learnt from a multimodal learning analytics deployment in-the-wild. *ACM Transactions on Computer-Human Interaction*, 31(1), 1-41.

Martínez-Maldonado, R., Yan, L., Deppeler, J., Phillips, M., & Gašević, D. (2022). Classroom Analytics: Telling stories about learning spaces using sensor data. In *Hybrid learning spaces* (pp. 185-203): Springer International Publishing.

Milesi, M. E., Alfredo, R., Echeverria, V., Yan, L., Zhao, L., Tsai, Y.-S., & Martinez-Maldonado, R. (2024). "It's Really Enjoyable to See Me Solve the Problem like a Hero": GenAI-enhanced Data Comics as a Learning Analytics Tool. Paper presented at the Extended abstracts of the CHI conference on human factors in computing systems.

Sha, L., Fincham, E., Yan, L., Li, T., Gašević, D., Gal, K., & Chen, G. (2023). The road not taken: preempting dropout in moocs. Paper presented at the International Conference on Artificial Intelligence in Education.

Shao, H., Martinez-Maldonado, R., Echeverria, V., Yan, L., & Gasevic, D. (2024). Data storytelling in data visualisation: Does it enhance the efficiency and effectiveness of information retrieval and insights comprehension? Paper presented at the Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems.

Yang, K., Raković, M., Liang, Z., Yan, L., Zeng, Z., Fan, Y., . . . Chen, G. (2025). Modifying AI, enhancing essays: How active engagement with generative AI boosts writing quality. Paper presented at the Proceedings of the 15th International Learning Analytics and Knowledge Conference.

Zhao, L., Echeverria, V., Swiecki, Z., Yan, L., Alfredo, R., Li, X., . . . Martinez-Maldonado, R. (2024). Epistemic network analysis for end-users: Closing the loop in the context of multimodal analytics for

collaborative team learning. Paper presented at the Proceedings of the 14th learning analytics and knowledge conference.

Zhao, L., Echeverria, V., Tan, Y., Yan, L., Li, X., Alfredo, R., . . . Osborne, A. (2024). Ordered Networked Analysis of Multimodal Data in Healthcare Simulations: Dissecting Team Communication Tactics.

Zhao, L., Gašević, D., Swiecki, Z., Li, Y., Lin, J., Sha, L., . . . Martinez-Maldonado, R. (2024). Towards automated transcribing and coding of embodied teamwork communication through multimodal learning analytics. *British Journal of Educational Technology*, 55(4), 1673-1702.

Zhao, L., Swiecki, Z., Gasevic, D., Yan, L., Dix, S., Jaggard, H., . . . Alfredo, R. (2023). METS: Multimodal learning analytics of embodied teamwork learning. Paper presented at the LAK23: 13th International learning analytics and knowledge conference.

Zhao, L., Tan, Y., Gašević, D., Shaffer, D. W., Yan, L., Alfredo, R., . . . Martinez-Maldonado, R. (2023). Analysing verbal communication in embodied team learning using multimodal data and ordered network analysis. Paper presented at the International Conference on Artificial Intelligence in Education.

Zhao, L., Yan, L., Gasevic, D., Dix, S., Jaggard, H., Wotherspoon, R., . . . Martinez-Maldonado, R. (2022). Modelling co-located team communication from voice detection and positioning data in healthcare simulation. Paper presented at the LAK22: 12th international learning analytics and knowledge conference.